Name:
Date: Thursday, December 12, 2019
PLSC 43401: Mathematical Foundations of Political Methodology
Final Exam

1. This exam has 6 sections.
2. There are a total of 100 possible points (excluding bonus question).
3. Answer as many questions as you can. You do not need to answer the questions in order. Try to answer the later parts of a question even if you have difficulty with earlier parts.
4. Please write your answers in the space provided in the exam.
5. Please clearly label your final answers where appropriate.
6. Do not take this exam with you.
7. If you have any clarification questions on the exam problems, feel free to ask.
8. Good luck!

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8 4 1) If $P(A \cap B) = P(A)P(B)$, which of the following statements are true? And explain why the statement is true/false briefly.

a) $P(A|B) = P(B)$

F. b/c $A \perp B$ $\cancel{A \perp B}$ $\Rightarrow P(A|B) = P(A)$

b) $P(A|B^c) = P(A)$

T b/c $A \perp B \Rightarrow A \perp B^c$

c) $P(A) + P(B) = P(A \cup B)$

F $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

d) $A \cap B = \emptyset$

F. Not necessarily.

5 4 2) Roll a fair four-sided die twice. What is the probability that the maximum of either roll is 2?

|Sample space| = 16

$\frac{3}{16}$

(1, 2), (2, 1), (2, 2)

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3) Suppose we have two sets of states, one with voter ID laws, one without. We have a bunch of state level characteristics, average income, percentage of white population, and electoral turnout. Our goal is to test whether each of these characteristics are statistically different than one another in our sample.

Suppose that even if the two groups of states are similar, there is some chance of a *false positive*, or type I error. Suppose that our test is set up so that we will only reject the null hypothesis of equality in average incomes, percentage of white population, with probability 0.1. For example, 1/10 of the time, we will falsely find that the two groups of states are different in income, even if they are not really different.

3 a) With no other information, what can we say about probability that we fail to detect a single difference in these three measures?

$\mathbb{P}(\text{reject } H_0 \mid H_0 \checkmark)$

H_0: no difference.

~~$\mathbb{P}(A_1^c \cup A_2^c \cup A_3^c)$~~

$\mathbb{P}(A \cup B \cup C) \leq P(A) + P(B) + P(C)$

$= 0.1 + 0.1 + 0.1$

$= 0.3$

$\mathbb{P}(A^c \cap B^c \cap C^c) = \mathbb{P}((A \cup B \cup C)^c) = 1 - \mathbb{P}(A \cup B \cup C) \geq 0.7$.

3 b) What if we knew that the percentage of the white population, turnout and average incomes were independent?

$\mathbb{P}(A^c \cap B^c \cap C^c) = \mathbb{P}(A^c) \cdot \mathbb{P}(B^c) \cdot \mathbb{P}(C^c)$

$= (1-0.1) \times (1-0.1) \times (1-0.1)$

$= 0.9^3$

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4) TRUE or FALSE. Also, explain why it is true/false briefly.

a) If X and Y are independent, $Var(X+Y) = Var(X) + Var(Y)$

T $= Var(X) + Var(Y) + 2cov(X, Y)$

b) $P(A|\Omega) = P(A)$

T $= \frac{P(A \cap \Omega)}{P(\Omega)} = \frac{P(A)}{1} = P(A)$

c) $P(A \cap B) \leq P(B|A)$

T $\frac{P(B|A)}{P(A \cap B)} = \frac{\frac{P(AB)}{P(A)}}{P(AB)} = \frac{1}{P(A)} \geq 1$ b/c $P(A) \in (0,1)$

d) $P(A \cup B) \leq P(A) + P(B)$

T $P(A \cup B) + P(A \cap B) = P(A) + P(B)$

e) $\bar{X} = E(X)$

F $\bar{X}$ is r.v.

II Election Night! 10

ABC is planning its election night coverage. The directors know that the audience doesn't care about individual races, just which party (Republicans or Democrats) will have control of the Senate. Calling the control of the US Senate is down to 5 key battleground states: Nevada, Missouri, North Dakota, Texas and Florida. ABC will call the race for the first party that wins three.

3 1) If both parties are equally likely to win, what is the probability that one party or the other sweeps, that is, wins all 5 races?

$$\binom{2}{1} \times 0.5^5 = 2 \times 0.5^5 = 0.0625$$

$$= \frac{1}{16}$$

The states are in four time zones and have variously efficient election systems, so each of the listed races come in one hour past each other, with Nevada at 7PM, Missouri at 8PM, North Dakota at 9PM, Texas at 10 PM, all the way to Florida at 11PM.

4 2) If both parties are equally likely to win, what is the probability that ABC will have to wait until 10PM?

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$$\mathbb{P}(\text{until } 10) = \mathbb{P}(9) + \mathbb{P}(10)$$

$$= \binom{2}{1} \times 0.5^3 + \binom{2}{1}\binom{3}{1} 0.5^4$$

$$= \frac{1}{4} + \frac{3}{8}$$

$$= \frac{5}{8}$$

3 3) What is the expected time to call control of the Senate?

$$\mathbb{P}(11) = \binom{2}{1} \times \binom{4}{2} 0.5^5 = \frac{6}{16} = \frac{3}{8}$$

$$= \frac{4 \times 3}{2}$$

$$\frac{1}{4} \times 9 + \frac{3}{8} \times 10 + \frac{3}{8} \times 11$$

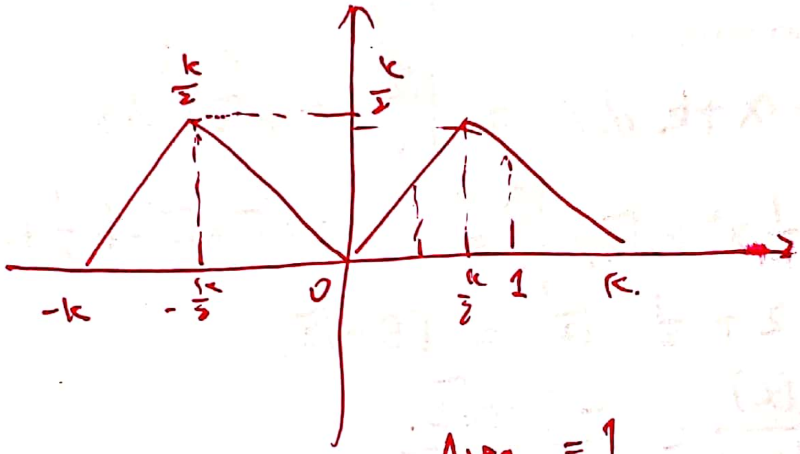
III Double Triangle Distribution 20

Suppose that X is a random variable distributed as follows:

$$f(x) = \begin{cases} x+k & \text{for } -k \le x \le -\frac{k}{2} \\ -x & \text{for } -\frac{k}{2} \le x \le 0 \\ x & \text{for } 0 < x \le \frac{k}{2} \\ -x+k & \text{for } \frac{k}{2} < x < k \end{cases}$$

5 1) What is k? [Hint: you can answer the second question first.]

$\sqrt{2}$



Area $= 1$

$$k \times \frac{k}{2} \times \frac{1}{2} \times 2 = 1$$

$$k = \sqrt{2}$$

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2) Draw f(x).

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3) Draw the conditional probability density function, $f_X(X|X>1)$.

$$P(X>1) = \int_1^k -x+k \, dx = \int_1^{\sqrt{2}} (-x+\sqrt{2}) \, dx$$

$$= -\frac{1}{2}x^2+\sqrt{2}x \Big|_1^{\sqrt{2}} = -\frac{1}{2}\times 2+\sqrt{2}\times\sqrt{2}-(-\frac{1}{2}+\sqrt{2})$$

$$= -1+2+\frac{1}{2}-\sqrt{2} = 1.5-\sqrt{2}.$$

$$f_X(x|X>1) = \frac{f_X(x)}{P(X>1)} = \frac{-x+\sqrt{2}}{1.5-\sqrt{2}}.$$

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4) Using your answer from above, if we know that X is greater than 1, what is the formula for the expectation of X? Note: you do not need to evaluate the integral.

$$\mathbb{E}[X|X>1] = \int_1^{\sqrt{2}} x \cdot f_X(x|X>1) \, dx$$

$$= \int_1^{\sqrt{2}} x \cdot \frac{(-x+\sqrt{2})}{1.5-\sqrt{2}} \, dx$$

IV Calculating Expectations 30

15 1) Suppose that X is a random variable distributed as follows:

$$f(x) = \begin{cases} 0 & \text{for } 0 \leq x < 3 \\ 6/x^2 & \text{for } 3 \leq x \leq 6 \\ 0 & \text{for } 6 < x \end{cases}$$

5 a) Verify f(x) is a valid pdf.

$$\int_3^6 \cancel{f(x)}\ 6/x^2\ dx$$

Also $f(x) > 0$

$$= \cancel{8}6 \times \left(-x^{-1}\right)\Big|_3^6$$

$$= -\frac{1}{6} + \frac{1}{3}$$

$$= \frac{1}{6}$$

$$= 1$$

5 b) What is E[X]?

$$\int x f(x) = 6 \int_3^6 \frac{1}{x}\, dx = 6 \times (\ln 6 - \ln 3)$$

$$= 6 \ln(2)$$

5 c) What is Var[X]?

$$E[X^2] = \int_3^6 6\,dx$$

$$= 6x\Big|_3^6 = 36 - 18 = 18$$

$$Var(X) = E[X^2] - E[X]^2 = 18 - \left[6 \times \ln(2)\right]^2$$

8 2) Suppose X and Y are random variables that are jointly distributed
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$$f(x,y) = \begin{cases} \frac{16y}{x^3}, \text{ if } x > 2, 0 < y < 1 \\ 0, \text{ otherwise} \end{cases}$$

5 a) What is the formula for $f(x)$? What is $f(y)$?

$$f_X(x) = \int_0^1 \frac{16y}{x^3}\, dy = \frac{1}{x^3} 8y^2 \Big|_0^1 = \frac{8}{x^3}$$

$$f_Y(y) = \int_2^{+\infty} \frac{16y}{x^3}\, dx = 16y \int_2^{+\infty} x^{-3}\, dx$$

$$\cancel{= \frac{1}{-4} x^{-2}} = \left(\frac{1}{-2} x^{-2} \Big|_2^{+\infty} \right) \times 16y$$

$$= 0 - \left(-\frac{1}{2}\right) \cdot \frac{1}{2^2} \times 16y$$

$$\cancel{= 4y}$$

$$= \frac{1}{8} 16y$$

$$= 2y.$$

5 b) Are X and Y independent?

$$f_X(x) f_Y(y) = f(x, y)$$

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4 c) What is E[Y]?

$$\int_0^1 2y \cdot y \, dy$$
$$= 2 \int_0^1 y^2 \, dy$$
$$= 2 \cdot \frac{1}{3} y^3 \Big|_0^1 = \frac{2}{3}$$

1 d) What is E[$Y|X = 3$]?

$$\parallel$$
$$E[Y]$$

V Inference 20

10 1) Suppose you are interested in the expectation of household income X_i (in dollars) of a population, based on n i.i.d samples $X_1, ..., X_n$ from the population. You use the sample mean $M_n = \frac{X_1+...+X_n}{n}$ to estimate $\mathbb{E}[X_i]$. In addition, you have a rough guess that $\sqrt{\text{Var}(X_i)}$ is \$100.

5 a) Give the formula for $\text{Var}(M_n)$.

$$= \frac{1}{n} Var(X_i)$$

5 b) How large should n be so that the standard deviation of M_n is at most 1\$?

$$\sqrt{Var(M_n)} \leq 1$$

$$Var(M_n) \leq 1$$

$$\frac{1}{n} Var(X) \leq 1$$

$$Var(X) \leq n$$

$$\parallel$$

$$10000$$

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2) Suppose you are interested in a random variable X with expectation $\mathbb{E}[X] = 55$ and population standard deviation $s = 3$. You draw a sample from the population and the sample size is 50 (i.e., $n = 50$). From the sample, you compute the sample mean: $\bar{X} = \frac{X_1+\ldots+X_{50}}{50}$.

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i. Using Chebyshev, what is the probability that the absolute value of the difference between sample mean and $\mathbb{E}[X]$ is greater than six?

$$\cancel{\mathbb{P}[|\mathbb{E}[X] - \bar{X}_n| \geq 6] \leq \frac{Var(\bar{X})}{}}$$

$$\mathbb{P}[|\bar{X} - \mathbb{E}[X]| \geq 6] \leq \frac{Var(\bar{X})}{6^2} = \frac{\frac{1}{n} Var(X_i)}{6^2}$$

$$= \frac{1}{50} \times \frac{9}{6 \times 6}$$

$$\cancel{\approx 0.006}$$

$$= 0.005$$

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ii. Given your previous answer, in expectation what is the largest number of observations that would fall outside the interval (49, 61)?

$$50 \times 0.005 = 0.250$$

$$= 0.25.$$

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VI BONUS: WILL TRUMP WIN 2020 ELECTION?

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(a) Before you use the Central Limit Theorem to construct a confidence interval, please state the Lindberg-Levy CLT.

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(b) Arthur is interested in whether or not Trump will win the 2020 presidential election. To answer this question, Arthur has collected his own data. Specifically, he designed a survey asking whether or not voters will vote for Trump in 2020 election, and constructs a random variable measuring support for Turmp. That is, for each respondent i, $X_i = 1$ if voter says he/she will vote for Trump, $X_i = 0$ if voter says he/she will not vote for Trump. Using a grant that he got from a legitimate US research organization, Arthur collects a sample with size n, where n is very large. Suppose the sample is i.i.d. and people respond honestly.

Suppose the fraction of voters vote for Trump is p, which is unknown to Arthur and everybody else. Therefore, we have: $\mathbb{E}[X_i] = p$ and $\text{Var}[X_i] = p(1-p)$. To estimate p, Arthur uses the following estimator: $M_n = \frac{X_1 + \ldots + X_n}{n}$. In addition, Arthur estimates $\text{Var}[X_i]$ by $M_n \times (1 - M_n)$. Arthur wants to use CLT to construct a 95% confidence interval for p, however, Arthur does not how to do it. Therefore, your task is to help Arthur **use CLT to construct a 95% confidence interval for p.** By the way, Arthur thanks for your help in advance.